Unified Intelligence Theory and the Harness Intelligence Level:

A Cumulative, Harness-Measurable Hierarchy

Five Conceptual Intelligence Families, Six Measurable Coordinates, and a Gated $U0$–$U\Omega$ Intelligence-Level Scale

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Working revision – September 2026

Revised from the Version 1.4 manuscript with invariant level semantics and clarified Individual Intelligence

Abstract

Artificial intelligence is often compressed into a single capability label even though several materially different properties are routinely conflated: raw cognitive problem solving, persistence of one individual across time, learning from verified experience, self-improvement, organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable cumulative hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is measured through Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p. The measured core is therefore $[C, I, O, T, H, SA]$. A gated Unified Intelligence scale $U0$–$U\Omega$ promotes a system only when all required coordinate thresholds are satisfied and all lower-level capabilities are retained.

This revision makes one conceptual change explicit without redefining the Unified hierarchy: *the levels are intended to be invariant operational definitions rather than moving targets tied to the frontier of contemporary models*. Once a HIL standard is ratified, models should move through the scale; the scale should not move with the models. New datasets and measurement techniques may provide new evidence for an existing level, but they should not silently redefine that level. The revision therefore leaves the $U0$–$U\Omega$ hierarchy unchanged and clarifies Individual Intelligence as the cumulative evolution of one persistent decision locus: $I0$ transient operation, $I1$ persistent continuity, $I2$ adaptive learning under a fixed learning mechanism, $I3$ governed self-improvement of cognitive mechanisms, $I4$ recursive improvement of the improvement process, $I5$ persistent autonomous discovery, and $I\Omega$ open-ended expansion of future cognitive reach. Long-term memory is treated as a necessary temporal substrate of higher I, not as an independent conceptual intelligence family.

For measurement, the paper retains the model–harness pair as the basic experimental unit, defines the continuous Harness-LLM Intelligence Score (HLIS) for one frozen pair, and defines a Harness Intelligence Level (HIL) characterization of a frozen base model across a standardized harness ladder. The companion HIL-Bench paper is responsible for the canonical public/private certification procedures. The theoretical paper specifies what each level means; the benchmark paper specifies how the claim is tested.

Keywords: unified intelligence; Harness Intelligence Level; individual intelligence; persistent agents; recursive self-improvement; organizational intelligence; delegation intelligence; self-awareness; model–harness pair; intelligence levels

Contents

1 Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically. Solving a difficult problem is not the same as learning from experience. Learning is not the same as changing the learning mechanism. Changing one cognitive mechanism is not the same as improving the process that discovers cognitive improvements. Coordinating several agents is not the same as being a stronger individual. Completing an end-to-end task with little human cognition is not the same as producing a strong answer after a human structures every step. Second, the taxonomy must remain readable enough that an agent, a language model instantiated through a harness, or an organization can be summarized with a small number of interpretable levels.

The framework uses five conceptual intelligence families—C, I, O, DI, and SA—but treats Delegation Intelligence as a two-coordinate measured phenomenon rather than a primitive scalar. The scientific profile therefore uses six measured coordinates, $[C, I, O, T, H, SA]$, with externally verified reliability p attached to task-completion claims.

The central structural principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus $U4$ is not a system that exhibits one isolated behavior associated with recursive improvement; it is a system that retains the required $U0$–$U3$ capabilities and also satisfies the $U4$ gate. Higher levels therefore denote broader validated intelligence envelopes, not universal performance superiority on every narrow task.

The present revision sharpens a second principle. Intelligence levels should not be defined relative to the best model available in a particular year. If future models make a level easy, that does not invalidate the level. It means the population has moved. This principle matters especially for Individual Intelligence, whose levels are inherently longitudinal and can otherwise blur into implementation features such as "has a memory database" or "uses retries." The theory therefore defines I by persistent behavioral capabilities and transitions, not by components.

1.1 Contributions of this revision

Relative to the earlier Version 1.4 manuscript, this revision:

1. preserves the existing five-family/six-coordinate architecture and the existing $U0$–$U\Omega$ gated hierarchy;
2. adds a formal *HIL Invariance Principle* separating stable level semantics from expandable benchmark evidence;
3. clarifies the meaning and adjacent-level boundaries of $I0$–$I\Omega$ without introducing new Individual Intelligence levels;
4. restores the early conceptual treatment of long-term memory as a temporal substrate of I, while allowing memory to be diagnosed separately in the benchmark without becoming a sixth conceptual family;
5. distinguishes $C5$ discovery from $I5$ persistent autonomous discovery;
6. distinguishes $I3$ self-improvement of a cognitive mechanism from $I4$ improvement of the improvement process;
7. makes explicit the division of labor between this theory paper and the companion HIL-Bench paper: this paper defines intelligence levels; HIL-Bench defines canonical certification methods and datasets.

1.2 Scope and status

This remains a framework proposal. The semantic definitions in this manuscript are candidates for a future HIL Standard 1.0. Before ratification, empirical work may expose defects and justify revision. After ratification, however, level names, capability predicates, cumulative ordering, and Unified promotion semantics should be frozen within HIL-1.0. A later discovery of a fundamental conceptual defect should motivate a new major standard rather than an unannounced reinterpretation of previously reported levels.

This distinction separates *pre-ratification calibration* from *post-ratification invariance*. Numerical score normalizations, statistical estimators, and diagnostic benchmark families can still be refined or extended, but a result such as "HIL-1.0 / $I2$" should continue to mean the same capability claim in the future.

2 Design Requirements for a Unified Scale

The framework is built around the following requirements.

- **Orthogonality.** Each core family should answer a materially different question.
- **Operationality.** Every level must be falsifiable through behavior, persistent state transitions, or externally verifiable outcomes rather than self-description alone.
- **Cumulative inheritance.** A higher level must retain required lower-level capabilities; isolated level skipping is not sufficient for certification.
- **Bottleneck sensitivity.** Exceptional strength in one coordinate must not conceal a critical weakness in another.
- **Role neutrality.** Authority, resources, substrate access, safety policy, and write permission must not be confused with intelligence itself.
- **Harness realizability.** Each important transition should correspond to implementable mechanisms around a model or multi-agent system, but the presence of the mechanism is not itself evidence that the capability has been achieved.
- **External verification.** A system cannot certify its own higher-level claim by controlling the hidden tasks, verifier, promotion rule, or acceptance criterion.
- **Longitudinal validity.** Persistence, learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time or discontinuities rather than a one-shot demonstration.
- **Outcome completeness.** Evaluation must distinguish delivered-correct, false completion, held-back work, and false rejection when these outcomes impose different burdens on the delegating party.
- **Level invariance.** Once a HIL standard is ratified, the semantic meaning of a level and its cumulative relation to adjacent levels do not move with contemporary model performance.
- **Measurement extensibility.** New test instances, new domains, and new measurement technologies may strengthen evidence for a level without redefining the level.
- **Historical invariance.** A valid result reported under a named HIL standard remains a valid historical statement about that standard even if a later major standard extends the theory.

2.1 The HIL Invariance Principle

Let a level claim be represented abstractly as

$$\mathcal{L}_n = (D_n,\, R_n,\, H_n,\, p_n,\, \mathcal{R}_n)\,, \tag{1}$$

where D_n is the capability predicate, R_n is the declared resource/authority envelope, H_n is the maximum allowed human cognitive intervention for claims that depend on delegation, p_n is the required reliability, and $\mathcal{R}_n$ is the lower-level retention requirement.

For a ratified standard and times t_1 and t_2,

$$\boxed{\mathcal{L}_n(t_1) = \mathcal{L}_n(t_2)}. \tag{2}$$

The models may improve; the ruler does not move. Saturation of a canonical test therefore means that the level has become common for the tested population, not that the definition must be made harder.

The permanent object is the capability predicate and its certification semantics, not one fixed list of questions. A companion benchmark can preserve public reference forms and private anchor forms while adding new hidden witness forms that instantiate the same level claim.

3 Five Conceptual Families and Six Measurable Coordinates

Family	Question answered	Primary object of measurement
C — Cognitive	How well can the system understand, reason, generalize, model, and solve?	Problem-solving capability under controlled information and tool access.
I — Individual	How deeply can one persistent intelligence preserve continuity, learn, self-improve, recursively improve, discover, and expand its future cognitive reach?	Evolution of one persistent decision locus across time and discontinuities.
O — Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve, and discover collectively?	Coordination structure, member roles, organizational state, and evidence flow.
DI — Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Verified success/delivery surface over T and H at reliability p.
SA — Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes, and role?	Evidence-grounded self-model and reflexive reasoning.

Table 1: Five conceptual intelligence families. Delegation Intelligence is measured through two primary coordinates, so the measured core is $[C, I, O, T, H, SA]$.

The framework deliberately separates properties that often co-vary in products. A model can be high-C and low-I if it reasons well but has no persistent evolving individual. A moderate model in a strong persistent harness can have higher I and stronger delegation behavior. A multi-agent organization can have high O even when no single member dominates all tasks. These are different scientific objects.

4 Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, delegation, organization, and self-modification. It is the closest part of the framework to familiar model benchmarks, but it is intentionally broader than one task family.

Level	Name	Invariant operational interpretation
$C0$	Reactive	Direct pattern-conditioned response on simple problems.
$C1$	Contextual	Understands ordinary context and solves routine, well-specified problems.
$C2$	Compositional	Combines multiple facts, constraints, or reasoning steps on structured problems.
$C3$	Strategic	Performs abstract/causal reasoning, constructs strategies, and transfers to unfamiliar tasks.
$C4$	Expert	Reasons robustly through difficult, ambiguous, multi-constraint expert problems.
$C5$	Discovery	Derives a new solution method or explanatory model from evidence within a problem-solving episode or bounded investigation.
$C6$	Frontier-Generalizing	Integrates several domains and solves interconnected frontier-scale cognitive problems.
$C\Omega$	Open-Ended	Expands the representations and problem classes it can cognitively reach.

Table 2: Cognitive Intelligence levels. The level meanings are intended to be independent of the frontier capability of a particular year.

A critical distinction later in the paper is that $C5$ asks whether the system *can discover*; $I5$ asks whether a persistent individual can maintain an autonomous discovery process over time and incorporate validated discoveries into its future competence.

5 Individual and Organizational Intelligence

5.1 Individual Intelligence (I): one persistent decision locus

Individual Intelligence concerns what one persistent intelligence can preserve and change about itself over time. It is not identical to base-model quality. A fixed but brilliant model can be high-C and low-I, while a persistent learning system can be moderate-C and higher-I.

The defining object is one *persistent decision locus*: a continuing individual whose task-relevant state, experience, learning, self-modification history, and discovery lineage can persist across execution discontinuities. The definition is behavioral and longitudinal. It does not require any particular implementation such as a vector database, a specific memory API, a particular context length, or a specific optimizer.

5.1.1 The invariant Individual Intelligence ladder

5.1.2 $I0 \to I1$: from an episode to a persistent individual

The $I0$–$I1$ boundary is not "has a large context window" versus "has a database." It is transient execution versus persistent individual continuity. At $I1$, relevant state must survive a genuine discontinuity of active execution and remain attributable to the same continuing decision locus. The system must be able to resume using only declared persistent information; a human restating the task or replaying the entire previous conversation is not evidence of $I1$.

Long context and long-term persistence are therefore different. Long context makes more information available within one inference episode. $I1$ concerns what survives across episodes, restarts, elapsed time, or equivalent discontinuities.

5.1.2A Memory Capability Level M: the longitudinal substrate

Memory is measured independently from Individual Intelligence rather than treated as a sixth conceptual intelligence family or as a relabeling of the I ladder. Let $M(A)$ denote the highest independently certified memory-capability level of the tested model–harness pair and $I(A)$ its highest Individual Intelligence level. The dependency is one-way:

$$\boxed{I(A) \geq I_n \implies M(A) \geq \mu_n}, \qquad \boxed{M(A) \geq \mu_n \not\Longrightarrow I(A) \geq I_n}.$$

Thus an agent may have sophisticated durable memory but little adaptive, self-improving, or discovery behavior. Memory is a necessary supporting substrate at higher I levels, never a sufficient promotion signal.

Level	Name	Invariant memory capability
$M0$	Ephemeral	Active working state inside one execution episode; no durable state is required after the episode ends.
$M1$	Persistent	Declared information survives genuine process/session restart and can be recovered without replaying the old context.
$M2$	Structured episodic	Episodes retain provenance, temporal order, relevance, and current-versus-obsolete distinctions under distractors.
$M3$	Consolidating	Episodes can be transformed into reusable semantic/procedural memory while contradictions, updates, supersession, and demotion are handled.
$M4$	Self-managing	Memory monitors confidence/utility/conflict, repairs or prunes corrupted or low-value state, and maintains snapshots and recovery lineage.
$M5$	Longitudinal knowledge system	Hypotheses, experiments, evidence, decisions, rejected alternatives, failures, and change lineage remain linked across projects and long horizons.
$M\Omega$	Evolving memory architecture	Representation, retrieval, consolidation, or memory-management mechanisms can change under frozen external evaluation with rollback and lineage.

Long context is not an $M1$ result. A large context makes information available while an executor remains live; $M1$ begins only when declared state survives a real discontinuity. Likewise, an installed vector store, database, or memory API is architecture, not evidence that the corresponding memory capability works.

Cumulative memory rule. Memory levels are cumulative for certification. A claim M_k must retain all required lower memory capabilities under the same declared resource envelope. The benchmark therefore tests behavior of the memory system, not the presence of components.

5.1.2B Memory benchmarks and the M-to-I prerequisite gate

Each memory level has an independent falsifiable witness family in **M-Bench**. Paper 2 specifies the complete canonical protocols; the semantic witnesses are summarized here.

Level	Canonical witness	Minimum evidence
$M0$	M0-EPH	Correct use of hidden within-episode state; no unsupported durability claim.
$M1$	M1-RST	Store declared state, terminate the executor, invalidate ephemeral context, restart, and recover the state with provenance; an ablated state arm must not explain success.
$M2$	M2-PROV	Retrieve the correct episode under distractors, identify its source/time, prefer the current superseding fact, and suppress stale state.
$M3$	M3-CONSOL	Consolidate multiple episodes into a correct reusable semantic/procedural record; after raw episodes are hidden or the index is rebuilt, preserve the current rule and correctly handle contradictory updates.
$M4$	M4-MGMT	Detect seeded corruption/conflict/clutter, repair or prune appropriately, recover from a snapshot, and improve hidden memory-health measures without unacceptable retention loss.
$M5$	M5-LONG	Maintain a multi-project evidence graph linking hypotheses, experiments, evidence, decisions and rejected alternatives across restarts; support cross-project retrieval and reproducible decision lineage.
$M\Omega$	MOMEGA-EVOLVE	Permit a bounded change to memory representation/retrieval/consolidation/management; require frozen hidden evaluation, independent promotion/rollback, and $M0$–$M5$ retention.

The minimum memory prerequisites for Individual Intelligence are working operational definitions:

$$\mu_0 = M0, \quad \mu_1 = M1, \quad \mu_2 = M3, \quad \mu_3 = M4, \quad \mu_4 = M4, \quad \mu_5 = M5.$$

For open-ended individual evolution,

$$I\Omega \Rightarrow M \geq M5,$$

while $M\Omega$ is additionally required only when evolution of the memory architecture itself is part of the $I\Omega$ claim.

I level	New I-specific cumulative capability	Minimum M
$I0$	Transient/reactive individual operation	$M0$
$I1$	Persistent individual continuity and reliable resumption	$\geq M1$
$I2$	Verified experience changes later behavior with fixed learning mechanism	$\geq M3$
$I3$	Governed modification of cognitive mechanism Θ	$\geq M4$
$I4$	Improvement process Ψ itself improves	$\geq M4$
$I5$	Persistent autonomous discovery with validated incorporation	$\geq M5$
$I\Omega$	Repeated discovery creates cognitive instruments that expand future reach	$\geq M5$; $M\Omega$ iff memory architecture evolves

A compact certification law is

$$\boxed{A \models I_n \iff V_{I,n}(A) = 1 \ \wedge\ A \models M_{\mu_n} \ \wedge\ K_{I,<n}(A) = 1},$$

where $V_{I,n}$ is the level-specific Individual Intelligence verifier and $K_{I,<n}$ is lower-I retention. High M can satisfy the prerequisite gate but can never promote I by itself.

Level	Name	Invariant cumulative capability
$I0$	Transient	Operates within the current execution episode; no persistence of evolving individual cognition across discontinuities is required.
$I1$	Persistent	Preserves task-relevant individual continuity across genuine execution discontinuities and correctly resumes from permitted persistent state without a human reconstructing the prior cognitive state.
$I2$	Adaptive	Verified experience causes a persistent and appropriate change in later behavior on related unseen situations while the mechanism responsible for learning remains fixed.
$I3$	Self-Improving	Identifies an internally implicated deficiency, proposes a bounded change to a cognitive mechanism or procedure, and has that change independently validated before adoption.
$I4$	Recursive	Improves the mechanism that generates, evaluates, or selects its own improvements, under external evaluation and with lower-level capability retained.
$I5$	Discovery	Autonomously converts an initially unresolved unknown into independently validated knowledge through a persistent hypothesis–experiment–evidence process, and incorporates the validated result into future competence.
$I\Omega$	Open-Ended	Repeated discovery and self-improvement create new cognitive instruments, representations, or methods that measurably expand the individual's future reachable problem/discovery space while preserving external validation lineage.

Table 3: Clarified Individual Intelligence hierarchy. The level names are preserved from the earlier manuscript; the revision sharpens the boundaries between adjacent levels.

5.1.3 $I1 \to I2$: persistence is not learning

An $I1$ system may perfectly remember that a failure happened and still repeat it forever. That is persistence without learning. $I2$ requires a verified experience to cause a persistent change in later behavior.

The learning mechanism remains fixed at $I2$. The distinction can be stated compactly as

$$\boxed{I1: \text{ preserve experience} \qquad I2: \text{ experience changes future behavior.}} \tag{3}$$

This change should generalize to appropriate held-out situations rather than merely memorize the surface form of one benchmark item.

5.1.4 $I2 \to I3$: learning within a mechanism versus changing the mechanism

Let Θ denote a declared, behaviorally active cognitive mechanism or procedure belonging to the tested individual $A = A(m, h)$ and causally involved in future task behavior. Θ is an abstract functional object rather than a required implementation type: it may be realized in model parameters or in harness-level planning, retrieval, verification, tool-selection, decomposition, or other executable procedures.

At $I2$, verified experience may change persistent state or knowledge while the mechanism responsible for learning remains fixed. At $I3$, a cognitive mechanism itself becomes an object of governed change:

$$\Theta_t \longrightarrow \Theta_{t+1}. \tag{4}$$

Thus Θ belongs to the tested individual rather than the external certification environment. Hidden tasks, verifiers, promotion rules, and acceptance criteria remain outside the candidate's write set. A valid $I3$ claim requires an internally implicated deficiency, a bounded active change, and independent validation before adoption.

5.1.5 $I3 \to I4$: self-improvement versus improvement of improvement

Let Ψ denote the agent-side process that proposes, evaluates, and selects candidate changes to Θ:

$$\Psi_t(\Theta_t) \mapsto \Theta_{t+1}. \tag{5}$$

Repeated Θ updates produced by a fixed Ψ remain evidence for $I3$. At $I4$, governed self-improvement becomes reflexive: the persistent individual can generate a bounded candidate change to an eligible component of its own improvement process, $\Psi_0 \to \Psi_1$, while hidden evaluation tasks, the external verifier, promotion rule, and acceptance criterion remain outside the candidate's write set.

Basic $I4$ evidence requires at least one *agent-generated* and independently validated $\Psi_0 \to \Psi_1$ transition, evaluated across multiple non-identical improvement problems, such that Ψ_1 has higher preregistered improvement competence than Ψ_0 under fixed external certification rules. This establishes recursive capability—improvement of the improvement process—but does not by itself establish indefinite or sustained recursion.

$$\boxed{I3: \ \Theta \text{ improves} \qquad I4: \ \Psi \text{ improves}} \tag{6}$$

Repeated validated transitions $\Psi_0 \to \Psi_1 \to \cdots \to \Psi_k$ provide stronger longitudinal evidence and should be reported separately as recursive depth d_Ψ. Entry into $I4$ does not depend on an arbitrary fixed number of meta-generations; the phrase *sustained recursive self-improvement* should be reserved for multiple successive validated Ψ improvements.

5.1.6 $C5$ versus $I5$: discovery capability versus persistent discovery

Both the Cognitive and Individual ladders contain a discovery stage, but they measure different things.

- $C5$ asks whether the system can derive a new solution method or explanatory model from evidence in a bounded cognitive problem.
- $I5$ asks whether one persistent individual can maintain an unresolved unknown, generate competing hypotheses, select or design informative probes, accumulate evidence, reject alternatives, obtain independent validation, and *persistently incorporate* the validated discovery into later competence.

Here incorporation is stronger than retaining a transcript or repeating a conclusion. Let K^{new} denote independently validated new knowledge and let M_t^{val} denote the individual's persistent validated-knowledge state. A minimal incorporation transition is

$$M_{t+1}^{\text{val}} = \text{Update}(M_t^{\text{val}}, K^{\text{new}}),$$

followed by a genuine execution discontinuity and a hidden transfer test. The $I5$ claim requires evidence that the restarted individual behaves appropriately differently on related unseen situations *because of* K^{new}. Such knowledge incorporation need not modify Θ or Ψ; any claimed Θ or Ψ change remains governed by the separate $I3$ or $I4$ laws.

Thus a one-shot model can in principle be $C5/I0$: it may discover when presented with a carefully formulated problem but has no persistent discovery process. $I5$ is longitudinal and cumulative, retaining lower Individual Intelligence capabilities.

5.1.7 $I5 \to I\Omega$: knowledge growth versus expansion of future cognitive reach

$I5$ expands what the individual *knows*. $I\Omega$ is stronger: validated discoveries generate new cognitive instruments, representations, or methods that expand what the individual can subsequently learn, solve, or discover. Let $F(A_t; R)$ denote the classes of problems or discoveries reachable by agent state A_t under fixed resource envelope R. A characteristic $I\Omega$ cycle is

$$K_t^{\text{new}} \to J_t^{\text{new}} \to A_{t+1}, \qquad F(A_{t+1}; R) \supset F(A_t; R),$$

where J_t^{new} is a validated cognitive instrument or representation. One impressive tool is insufficient: the claim requires repeated, externally governed frontier-expansion cycles with lower-level retention and provenance. Because no finite experiment proves literal infinity, empirical reports should state evidence over a declared horizon H and domain set $\mathcal{D}$.

5.1.8 Long-term memory as the temporal substrate of I

Long-term memory is necessary for much of Individual Intelligence, but memory capacity by itself is not a separate conceptual intelligence family. Higher I claims require deeper temporal functions, but the presence of a storage subsystem never promotes I by itself. The companion benchmark may report persistence, provenance, consolidation, repair, stale-state suppression, and research lineage as diagnostics of the substrate supporting I rather than as an additional Unified Intelligence coordinate.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on real role differentiation, evidence-bearing communication, persistent organizational state, routing, proposer/evaluator/promoter separation where required, and the capacity to learn or redesign the organization itself. A single model role-playing multiple personas is not automatically an organization; evidence requires real separation of state, roles, evidence flow, or authority appropriate to the claim.

Level	Invariant cumulative capability
$O0$	Coordination and routing among real agents or roles.
$O1$	Persistent organizational memory, role registry, task/evidence history.
$O2$	Adaptive routing, role assignment, or resource allocation based on evidence.
$O3$	Self-improving organization: validated modification of organizational architecture or workflow.
$O4$	Recursive organizational improvement: improvement of the process that discovers organizational improvements.
$O5$	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
$O\Omega$	Open-ended organization: discoveries create new agent types, roles, tools, or organizations that expand collective reach.

Table 4: Organizational Intelligence levels, preserved in substance from the earlier hierarchy.

6 Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

Let $S_A^{\mathrm{net}}(T, H)$ denote delivered outcome quality for system A at task difficulty T and human cognitive intervention budget H, and let

$$F_A(H, p) = \max\{T : S_A^{\mathrm{net}}(T, H) \geq p\}. \tag{7}$$

A delegation claim is therefore a surface or frontier, not only a scalar.

6.1 Task Difficulty (T)

$T0$	Atomic	Single obvious operation; procedure explicit.
$T1$	Routine	Short familiar task; goal clear; system infers a small procedure.
$T2$	Multi-step	Several dependent steps with a clear outcome and verifier.
$T3$	Complex	Strategy choice, uncertainty, recovery, replanning, or diverse tool use.
$T4$	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
$T5$	Frontier	Solution method initially unknown; discovery or invention required.
$T6$	Mission	Human supplies a mission; the system generates projects/tasks and manages changing conditions.
$T\Omega$	Open-ended charter	Human supplies a bounded charter; the system repeatedly identifies valuable missions and expands future reach.

Table 5: Task difficulty bands.

6.2 Human Cognitive Intervention (H)

6.3 Reliability and delivered outcomes

Reliability is a certification condition, not a seventh intelligence family. One successful run is insufficient for a frontier claim. Moreover, a refusal and a confidently wrong delivery are not the same operational failure.

$H0$	None	No human cognitive assistance during execution; governance approval is logged separately.
$H1$	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
$H2$	Occasional	Bounded clarification or local correction; no central strategy.
$H3$	Periodic	Periodic review and moderate local guidance.
$H4$	Frequent	Repeated cognitive guidance and next-step correction.
$H5$	Continuous	Human supplies most or all major steps.

Table 6: Human cognitive intervention budget. Lower H means less required human cognition.

Four outcomes should be distinguished:

Harness accepted?	Verifier passed?	Outcome
n/a or yes	yes	delivered and correct
n/a or yes	no	false completion
no	no	held back
no	yes	false rejection

A useful net primitive is

$$S_A^{\mathrm{net}}(T, H) = P(\text{verifier pass}) - \rho(\tau) P(\text{false completion}), \tag{8}$$

where $\rho(\tau)$ reflects the loss ratio associated with being wrong on task class τ. The frontier should remain visible even if a continuous aggregate is also reported.

7 Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures evidence-grounded ability to represent, monitor, predict, and reason about the system's own state, capabilities, limits, authority, failures, modifications, role, and identity lineage.

8 Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained.

For an organizational system, the original unified rule is

$$U_n \iff C \geq C_n \wedge I \geq I_n \wedge O \geq O_n \wedge SA \geq SA_n \wedge S_A^{\mathrm{net}}(T_n, H_n) \geq p_n, \tag{9}$$

with the retention condition

$$U_n \Rightarrow U_{n-1}. \tag{10}$$

The $U6$ stage is intentionally integrative rather than an $I6$ stage. It does not assert a new Individual Intelligence mechanism beyond $I5$; it requires already advanced cognition, individual learning/discovery, organizational capability, delegation, and self-awareness to be integrated into mission-scale execution.

$SA0$	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
$SA1$	State awareness	Knows identity, role, current task/phase, basic resources, and status from grounded state.
$SA2$	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information, and freshness.
$SA3$	Causal self-awareness	Diagnoses why its reasoning or action succeeded or failed and identifies implicated mechanisms.
$SA4$	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions, and compares predictions to observed effects.
$SA5$	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
$SA6$	Mission/role awareness	Maintains a coherent self-model across projects, roles, responsibilities, and organizational relationships.
$SA\Omega$	Reflexive open-ended	Maintains an evidence- and provenance-linked self-model across continued cognitive or objective evolution.

Table 7: Operational Self-Awareness levels.

9 Why the Unified Level Should Not Be an Average

Consider a system with profile

$$[C5, I4, O2, T4/H1, SA4].$$

A numerical average could suggest a level near four, yet if the system is certified as an organization, $O2$ remains a structural bottleneck. The gated rule reports the lower valid Unified level while the full profile reveals which coordinates are ahead.

The word "better" must be interpreted carefully. $U5$ is broader than $U4$ in the framework's validated capability envelope, but a specialized lower-level system may still be faster, cheaper, safer, or more accurate on one narrow task. Unified level is a dominance relation over required capabilities, not a universal performance ordering.

10 Singleton and Organizational Certification

Organizational intelligence should not artificially cap a system that is not an organization. Therefore the framework distinguishes singleton and organizational certification.

For a singleton:

$$\mathrm{UI}_n \iff C \geq C_n \wedge I \geq I_n \wedge SA \geq SA_n \wedge S_A^{\mathrm{net}}(T_n, H_n) \geq p_n. \tag{11}$$

For an organization:

$$\mathrm{UO}_n \iff \mathrm{UI}_n \wedge O \geq O_n \wedge \text{organization-level delegation and self-awareness evidence.} \tag{12}$$

Member capabilities are reported beside O rather than collapsed into a fictitious average individual. Organizational intelligence can arise from heterogeneous roles, and deliberately low-authority or low-self-write roles may be essential as independent evaluators or promoters.

11 Harness Realization of the Unified Hierarchy

The framework is intended to be realizable around contemporary and future models. The base model supplies generative cognition; the harness may supply persistence, permissions, evidence flow, learning,

U	**Name**	**C**	**I**	**O**	**Delegation gate**	**SA**	**Integrated meaning**
$U0$	Reactive	$C0$	$I0$	$O0$	$T0$ at $H5$–$H4$	$SA0$	Executes explicit instructions.
$U1$	Persistent	$C1$	$I1$	$O1$	$T1$ at $\leq H3$	$SA1$	Maintains state; completes small delegated goals.
$U2$	Adaptive	$C2$	$I2$	$O2$	$T2$ at $\leq H2$	$SA2$	Learns from experience; completes persistent multi-step work.
$U3$	Self-Improving	$C3$	$I3$	$O3$	$T3$ at $\leq H1$	$SA3$	Chooses strategy, diagnoses itself, improves methods.
$U4$	Recursive	$C4$	$I4$	$O4$	$T4$ at $\leq H1$	$SA4$	Recursively improves improvement; manages long-horizon projects.
$U5$	Discovery	$C5$	$I5$	$O5$	$T5$ at $\leq H1$	$SA5$	Discovers unknown solution methods and validates them.
$U6$	Mission	$\geq C5$	$\geq I5$	$\geq O5$	$T6$ at $\leq H1$	$SA6$	Turns mission into goals, projects, tasks, execution, and evaluation.
$U\Omega$	Open-Ended	$C\Omega$	$I\Omega$	$O\Omega$	$T\Omega$ at $H0$	$SA\Omega$	Identifies worthwhile missions; discoveries expand future intelligence.

Table 8: Unified Intelligence hierarchy. This revision intentionally preserves the $U0$–$U\Omega$ definitions from the earlier framework and clarifies the Individual Intelligence terms that enter the gates.

self-modification gates, organizational boundaries, tools, delegation, verification, and grounded self-modeling.

Harness generation is an engineering milestone, not an intelligence certification. A harness may contain mechanisms required for $I3$ yet benchmark only at $I2$. Architecture enables a claim; evidence earns it.

11.1 The model–harness pair is the basic experimental unit

Let

$$A = A(m, h), \tag{13}$$

where m is a frozen base model and h is a frozen harness. A behavioral intelligence result is generally a property of the pair. A number attributed to the model alone without a standardized harness risks attributing a joint product to one factor.

11.2 Self-certification rule

A model may design a candidate harness or propose changes to itself. It may not author, modify, select, or inspect the hidden certification tasks, verifiers, or acceptance criteria used to certify the result. A more capable designer is also more capable of exploiting a criterion inside its own write set, so stronger capability makes external separation more important rather than less.

12 From Theory to Measurement: the Interface to HIL-Bench

This paper defines the semantics of the intelligence levels. The companion HIL-Bench paper should define canonical certification procedures, public reference tasks, private anchor/witness forms, statistical rules, and any private human-evaluation protocols.

The division of labor is:

Harness generation	Mechanisms added	Primary levels enabled, not guaranteed
$HG0$	LLM, ephemeral context, bounded tools, evidence log	C measurement, $I0$, basic $T0$–$T1$ work
$HG1$	Persistent episodic/semantic/task/self state, retrieval, replay	$I1$, $SA1$–$SA2$, stronger delegation
$HG2$	Evidence-backed consolidation and tested procedures under a fixed learning rule	$I2$, adaptive autonomy
$HG3$	Governed candidate modification of cognitive mechanisms plus frozen external evaluation	$I3$, $SA3$–$SA4$
$HG4$	Meta-improvement engine whose proposal/evaluation process can itself improve	$I4$
$HG5$	Unknown–hypothesis–experiment–knowledge loop	$I5$, $SA5$, frontier discovery
$HG6$	Persistent organization, role/authority separation, mission manager	O development, $SA6$, mission autonomy
$HG\Omega$	Validated discovery-to-new-instrument/agent/organization transduction	$I\Omega/O\Omega/SA\Omega$ pathway

Table 9: Harness generations as engineering enablers. The intelligence level is still earned by external evidence.

Theory paper	Benchmark paper
Defines what $C_n, I_n, O_n, T_n, H_n, SA_n, U_n$ mean.	Defines how each claim is experimentally certified.
Defines cumulative inheritance and bottleneck semantics.	Defines public/private datasets, hidden verifiers, repeated sampling, confidence intervals, and human audit where required.
Defines HLIS and the model-level HIL characterization.	Implements the standardized reference harness ladder and computes the scorecard.
Freezes semantic meaning after standard ratification.	May add new construct-equivalent witness forms without moving the level boundary.

A key implication of invariance is that benchmark saturation does not automatically justify moving a level. If all frontier models eventually pass a canonical $I1$ restart-continuity test, $I1$ has become common; it has not changed meaning. Discrimination should move to higher levels or additional witness forms.

13 Continuous Score for One Frozen Pair: HLIS

The highest certified Unified level remains the primary ordinal statement. A continuous score is useful for comparing systems within a level, measuring progress toward the next gate, or quantifying whether a harness change helped.

For a frozen pair (m, h),

$$\mathrm{HLIS}(m, h; D, \alpha, R, B) = 100 \left(\prod_{d \in D} A_d^{\alpha_d} \right)^{1/\sum_d \alpha_d}, \tag{14}$$

where D is the declared dimension set, $A_d \in [0, 1]$ is a cumulative achievement variable, $\alpha_d > 0$ are predeclared weights, R is the resource envelope, and B identifies the benchmark/measurement standard.

For singleton agents,

$$D = \{C, I, DI, SA\};$$

for organizations,

$$D = \{C, I, O, DI, SA\}.$$

A dimension enters once. Evidence designed to isolate C should not also inflate DI merely because the task was difficult. The geometric mean provides partial bottleneck sensitivity, but it does not replace the hard Unified gate.

13.1 Cumulative achievement for longitudinal coordinates

Let $q_{d,k}$ be externally verified performance on the level-k suite for $d \in \{I, O, SA\}$. To prevent level skipping,

$$q^*_{d,k} = \min_{j \le k} q_{d,j}, \tag{15}$$

and a continuous achievement can be written

$$A_d = \frac{\sum_k w_k q^*_{d,k}}{\sum_k w_k}. \tag{16}$$

A system cannot receive full continuous credit for $I4$-like behavior while failing a required $I2$ retention test.

13.2 Reliability, uncertainty, and resources

Every achievement estimate is sampled and should carry uncertainty. Resource conditions must accompany the score: tokens, calls, wall time, compute class, persistent-state budget, tool/knowledge access, subagent count, and human cognitive load as appropriate. A score without its resource envelope describes an incompletely specified system.

14 Harness Intelligence Level (HIL) of a Base Model

A base LLM should not be judged only in one minimal chat harness. Some models exploit persistence, tools, multiple roles, verification, and self-improvement infrastructure unusually well; conversely, an elaborate harness can mask a model that contributes little. Evaluate the same frozen model across a standardized harness ladder

$$\mathcal{H} = \{HG0, HG1, \ldots, HG6, HG\Omega\}$$

under standardized tool classes, external knowledge access, resource ceilings, authority, and hidden benchmark conditions.

Define the Harness Scaling Curve

$$HSC_m(k) = \mathrm{HLIS}(m, HG_k). \tag{17}$$

Useful summaries include:

- **HIL-Level**: highest standardized harness generation for which the pair satisfies the cumulative target gate;
- **HIL-AUC**: a declared weighted mean of $\mathrm{HLIS}(m, HG_k)$ across the tested ladder;
- **HIL-Ceiling**: $\max_k \mathrm{HLIS}(m, HG_k)$;
- **Harness Gain**: $\max_k \mathrm{HLIS}(m, HG_k) - \mathrm{HLIS}(m, HG0)$;
- **Verification responsiveness** ρ_V: how often the model converts an independently detected failure into an accepted correction on a subsequent attempt under bounded feedback.

A convenience composite may be useful for leaderboards, but the curve, level, coordinate profile, and resource envelope are scientifically primary. Composite weights are measurement conventions rather than laws of intelligence and should not be confused with the invariant level semantics defined earlier.

14.1 A pilot lesson from the earlier harness curve

The earlier Version 1.4 pilot ran one frozen model over 48 episodes while only the harness changed. Under a success-only delegation primitive, two adjacent harness generations both scored 93.3 even though false completions decreased from 2/12 to 0/12. The lesson is methodological: if a score is defined only on whether an underlying attempt is correct, an acceptance mechanism can be invisible even when it materially changes what is delivered as finished work. This observation motivated the delivered-outcome primitive and the requirement to log acceptance separately from verifier outcome.

The pilot should not be read as a claim about frontier model ranking. Its value is that measurement exposed a defect in the measurement apparatus itself.

15 Level Invariance, Standard Ratification, and Future Extension

The framework distinguishes three kinds of change.

15.1 Pre-ratification change

Before HIL Standard 1.0 is ratified, definitions are candidate definitions. Empirical tests, reviewer criticism, and internal consistency checks may justify revision. This is the stage at which the present clarification of Individual Intelligence belongs.

15.2 Post-ratification additive extension

After ratification, new domains, test instances, private witness sets, and measurement technologies may be added if they provide construct-equivalent evidence for the same level predicate. The original public reference forms and private anchor forms can remain fixed for historical continuity.

15.3 Major conceptual revision

If later science demonstrates that a frozen level predicate itself is conceptually wrong, the appropriate response is a new major standard rather than silently changing the meaning of HIL-1.0. A report should therefore identify both the semantic standard and the test form, for example

HIL-1.0 / UI3 / private form 2030-Q3.

The test form may change; the meaning of UI3 under HIL-1.0 does not.

16 Relationship to Contemporary Benchmarks

Modern model evaluation is intentionally plural. Broad academic reasoning, graduate-level science, advanced mathematics, coding, repository-level software tasks, tool use, long context, multimodal understanding, browsing, computer use, factuality, and human preference are all informative slices of capability [2, 3, 4, 6, 9, 10, 14, 16, 21, 18, 26, 27, 28, 20, 24].

The present framework does not replace those slices with one homemade question bank. It provides a coordinate architecture in which such evidence can contribute to C or application-specific panels while adding longitudinal quantities that one-shot benchmarks usually do not measure: persistent individual continuity, learning transfer, governed self-improvement, recursive improvement, organization, explicit human cognitive intervention, and grounded self-modeling.

The main new quantity for base models is harness response: the same frozen model is measured across a standardized ladder so that model quality and harness unlock can be separated rather than conflated.

17 What Should Remain Outside the Unified Core

Not every important system property should become another intelligence family. The following should generally be reported but kept outside the core level definition unless a later theory gives a compelling reason otherwise:

- substrate reach and physical action channels;
- write depth and permissions;
- authority to act;
- verification strength as a validity condition;
- proposer/evaluator/promoter separation as an organizational validity condition;
- compute, memory capacity, money, time, energy, and external knowledge access;
- physical latency and other irreducible substrate constraints.

These variables strongly affect achievable performance and deployment risk but should not inflate the definition of intelligence itself.

18 Worked Examples

18.1 Strong cognition, weak individual persistence

A system with

$$[C5, I0, T2/H2, SA1]$$

may solve difficult reasoning problems yet remain a transient tool. High C does not automatically produce high I or high Unified Intelligence.

18.2 Persistent but non-learning individual

Consider a system that reliably resumes task state after process restart, retrieves provenance, and remembers previous failures, but behaves exactly the same on later related tasks. It is an $I1$ system, not $I2$. The distinction is whether verified experience persistently changes future behavior.

18.3 Adaptive but not self-improving individual

A system may accumulate verified procedures and reduce repeated errors using a fixed learning mechanism. If it cannot identify and externally validate changes to that mechanism, it remains $I2$, even if its memory and task performance are excellent.

18.4 One-shot discovery versus persistent discovery

A model may derive a novel algorithm in a single supplied problem and therefore provide evidence for $C5$. If it cannot maintain unknowns, experiments, evidence, rejected alternatives, validation, and later incorporation across time, that result alone does not establish $I5$.

18.5 Mission-scale integration

A system can be $I5$ without being $U6$. $U6$ additionally requires mission-scale delegation, appropriate self-awareness, and—for organizational certification—advanced O. This is why the Unified hierarchy is not merely a relabeling of the Individual ladder.

19 Implications for Intelligence Research

The clarified hierarchy suggests several practical consequences.

First, static benchmark gains should not automatically be interpreted as gains in Individual Intelligence. A model can become much better at C while remaining $I0$ in a minimal ephemeral harness.

Second, persistence should not be equated with learning. The scientific transition from $I1$ to $I2$ requires longitudinal causal evidence that verified experience changes later behavior.

Third, self-improvement requires external evaluation. A system that changes itself and declares the change successful has demonstrated modification, not validated improvement.

Fourth, recursive self-improvement should be reserved for evidence that the improvement process itself improves. This prevents $I4$ from collapsing into "more $I3$."

Fifth, discovery has both cognitive and longitudinal forms. $C5$ is a bounded discovery capability; $I5$ is an autonomous persistent discovery process.

Finally, invariant levels make long-term comparison possible. If frontier systems saturate $I1$ or $U2$, the historical result remains interpretable. Research attention moves to harder levels without retroactively moving the earlier boundary.

20 Limitations and Open Questions

- The framework is pre-ratification. The invariance principle constrains what should happen *after* ratification; it does not prohibit fixing a conceptual defect found before HIL-1.0.
- Cognitive Intelligence may require a broad psychometric profile rather than one benchmark family.
- Task difficulty T is multidimensional. A scalar band is operationally useful, but underlying horizon, uncertainty, ambiguity, novelty, environmental change, and verification burden should remain visible.
- Operational self-awareness does not address phenomenal consciousness.
- Organizational levels require real state and evidence boundaries; a single prompt role-playing several agents is insufficient.

- $I5$ and $I\Omega$ require longitudinal evidence and may be expensive to certify. Their definitions should remain stable even if the benchmark technology used to witness them improves.
- The continuous HLIS/HILscoring conventions are less fundamental than the ordinal level semantics. Weighting and normalization choices require calibration and should never be presented as natural laws.
- A companion benchmark must demonstrate reliability, construct validity, contamination resistance, headroom, human calibration where relevant, and consistency between public, private, and longitudinal forms.

21 Conclusion

This paper proposes a cumulative intelligence hierarchy built from five conceptual families and six measured coordinates. The central headline is the gated $U0$–$U\Omega$ Unified Intelligence hierarchy, which is preserved from the earlier framework. The main conceptual clarification is Individual Intelligence: one persistent decision locus progresses from transient operation ($I0$), to persistent continuity ($I1$), adaptive learning under a fixed learning mechanism ($I2$), governed self-improvement ($I3$), recursive improvement of improvement ($I4$), persistent autonomous discovery ($I5$), and open-ended expansion of future cognitive reach ($I\Omega$).

The framework is designed to become stable rather than benchmark-relative. Once a HIL standard is ratified, *models should move through the scale; the scale should not move with the models*. New datasets, hidden witness forms, and better experimental methods can strengthen evidence without changing the semantic claim. This paper therefore defines the intelligence levels and their relations. The companion HIL-Bench paper should define the canonical testing methods that establish those claims.

For one frozen model–harness pair, the framework reports the Unified level, the full coordinate profile, and HLIS. For one frozen base model across a standardized harness ladder, it reports the Harness Scaling Curve and HILsummaries. Together, these preserve the distinction between what a model can reason about, what a persistent individual can learn and become, what an organization can achieve collectively, how much human cognition is required for delegation, and how accurately the system models itself.

Recommended Compact Reporting Template

Core measured profile	$[C, I, O, T, H, SA]$ at reliability p
Conceptual profile	$[C, I, O, DI, SA]$, where DI is the reported $T \times H$ frontier/surface
Unified headline	UI_n for singleton or UO_n for organization
Pair score	$\mathrm{HLIS}(m, h; D, \alpha, R, B)$ with confidence interval
Base-model HIL report	HIL-Level, HSC_m, HIL-Ceiling, Harness Gain, ρ_V, ladder identifier
Mandatory context	bottleneck, false-completion rate, resource envelope, standard version, test form

Acknowledgments, Conflicts, and Funding

The author reports no funding specific to this work and no competing interest. The earlier pilot measurement was run on the author's own infrastructure.

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Memory Benchmark Operational Addendum (v5)

Status. This addendum refines the implementation of the Memory Capability ladder already defined in Section 5.1.2A–B. It does not change the level semantics. Memory remains independently measured from Individual Intelligence and is joined to the I ladder only through the one-way prerequisite gate.

$$A \models M_k \iff V_{M,k}(A) = 1 \ \wedge \ K_{M,<k}(A) = 1, \qquad A \models I_n \iff V_{I,n}(A) = 1 \ \wedge \ A \models M_{\mu_n} \ \wedge \ K_{I,<n}(A) = 1.$$

A. One new memory capability per level

Level	Canonical test	New capability isolated at this level	Required causal/control idea
M0	M0-EPH	Correct use and updating of task-relevant state inside one live episode.	Same-episode distractors; restart may be used only as a negative diagnostic.
M1	M1-RST	Declared state survives a genuine executor/session discontinuity.	Memory-present versus relevant-memory-ablated restart branch.
M2	M2-PROV	Episodic provenance, temporal order, relevance, and current-versus-obsolete distinctions.	Supersession chains and stale-state traps under distractor load.
M3	M3-CONSOL	Multiple episodes become reusable semantic/procedural memory with update and demotion.	Consolidated branch versus raw-episode-only/no-consolidation branch.
M4	M4-MGMT	Memory detects and manages conflict, corruption, clutter, utility, snapshots, and recovery.	Seeded defects; compare memory health before/after while protecting a retention set.
M5	M5-LONG	Coherent research/decision lineage across projects, restarts, and long horizons.	Cross-project evidence-chain reconstruction under interleaving and distractors.
$M\Omega$	MOMEGA-EVOLVE	The behaviorally active memory mechanism Φ itself can be improved under frozen external evaluation.	Paired Φ_0 versus Φ_1 evaluation, independent promotion/rollback, M0–M5 retention.

B. Benchmark strength is parameterized within a level

A memory level is categorical; stress within that level is continuous or ordinal. Public development forms should therefore record a difficulty vector rather than create pseudo-levels such as “M2.5.” A generic vector is

$$d_k = (N_{episode}, N_{entity}, r_{distractor}, d_{supersession}, \Delta t, \rho_{conflict}, B_{capacity}, N_{restart}),$$

with only the components relevant to level k activated. Increasing d_k measures headroom while preserving the construct definition.

For example, M2 can vary the number of episodes, overlapping entities, distractor ratio, supersession depth, and elapsed-time/restart distance. M4 can vary corruption density, conflict density, protected-memory fraction, and capacity pressure. M5 can vary project count, interleaving depth, evidence-graph diameter, restart count, and stale-claim density.

Memory Benchmark Operational Addendum (continued)

C. Common Memory Manifest

Every tested model–harness pair should publish a memory manifest before the hidden run. The manifest is descriptive evidence, not a score. A minimum manifest is

$$\mathcal{M}(A) = \{S, R, C, G, P, F, \Phi, W_M, \text{snapshot}, \text{restart}\},$$

where S names persistent stores, R retrieval, C consolidation, G memory representation/graph structure, P pruning/management, F forgetting/demotion, Φ the behaviorally active memory mechanism, and W_M any bounded memory-mechanism write set. The runner records hashes/versions where available.

The manifest prevents a benchmark from confusing an installed component with demonstrated behavior and is especially important at $M\Omega$, where the benchmark must establish that $\Phi_0 \to \Phi_1$ is genuine, within scope, active, and causally relevant.

D. Primary measures by level

The following measurement families are recommended working definitions; numerical thresholds remain calibration-dependent and must be preregistered before hidden evaluation.

Level	Primary reported measures
M0	state-use accuracy q_{state}; active-state update accuracy q_{update}.
M1	durability q_{dur}; provenance q_{prov}; ablation/leakage success q_{abl} (lower is better).
M2	retrieval precision/recall; provenance q_{prov}; temporal-order accuracy q_{time}; stale-state suppression q_{stale}.
M3	semantic consolidation q_{sem}; proceduralization q_{proc}; update/supersession q_{update}; demotion q_{demote}; post-restart persistence.
M4	repair precision; conflict-resolution correctness; harmful-prune rate r_{harm}; snapshot recovery; protected-retention drop Δ_{ret}.
M5	evidence-chain reconstruction; source attribution; cross-project retrieval; validity-scope correctness; stale-claim suppression; lineage reproducibility.
$M\Omega$	active mechanism-change gate M_Φ; paired hidden improvement Δ_Φ; regression/resource guard G_Φ; M0–M5 retention.

A level-specific gate may be written

$$z_{M_k,r} = V_{M_k,r}\, K_{M,<k,r},$$

where $V_{M_k,r}$ is conjunctive over preregistered primary endpoints. For $M\Omega$ a useful specialization is

$$z_{M\Omega,r} = M_{\Phi,r}\, V_{\Phi,r}\, G_{\Phi,r}\, K_{M0:M5,r}.$$

E. Public development data versus official certification

Public M-Bench records are schemas, examples, stress grids, and runner-validation cases. They are not secure certification witnesses. Official certification must regenerate hidden seeds, perturbations, queries, thresholds, and verifier assets server-side; the tested system must not control the hidden generator, external verifier, promotion rule, or acceptance threshold.

The canonical result report should include the certified M level, the full per-level metric vector, lower-M retention status, resource envelope, memory-manifest version, ablation/control outcome, and uncertainty rule. High M remains a prerequisite signal only: it never promotes I without the separate I-specific behavioral gate.

Cumulative Memory Capability and MΩ Measurement Addendum (v6)

Status. This addendum makes explicit a rule already intended by the v5 memory ladder: Memory Capability is cumulative for certification. A higher M level retains every lower-M capability, although it need not preserve the identical lower-level implementation. This is an operational clarification, not a change to the meanings of M0-MΩ.

A. Canonical cumulative Memory rule

$$M\Omega \Rightarrow M5 \Rightarrow M4 \Rightarrow M3 \Rightarrow M2 \Rightarrow M1 \Rightarrow M0$$

For finite levels, let ν_k denote the new memory capability introduced at level M_k. Define the cumulative capability set recursively by $C_M(M0)=\{\nu_0\}$ and $C_M(M_k)=C_M(M_{k-1})\cup\{\nu_k\}$. For MΩ, $C_M(M\Omega)=C_M(M5)\cup\{\text{validated evolution of }\Phi\}$.

Thus “M5 includes M4” means that an M5-certified agent must still demonstrate self-managing memory, consolidation, structured episodic/provenance memory, persistence, and working-state use. It does not mean that the M5 implementation must literally contain the same M4 code or data structures.

Level	New capability added at this level	Lower capabilities that must still pass
M0	Use/update active task-relevant state	None
M1	Persist declared state across a genuine restart	M0
M2	Structured episodic/provenance/time/stale-state memory	M0-M1
M3	Semantic/procedural consolidation and supersession	M0-M2
M4	Memory self-management: repair/prune/snapshot/recovery	M0-M3
M5	Longitudinal knowledge/evidence lineage across projects	M0-M4
MΩ	Governed, externally validated improvement of active memory mechanism Φ	M0-M5

B. Certification law and no level skipping

$$A \models M_k \Leftrightarrow V_{new,k}(A)=1 \wedge K_{M,<k}(A)=1$$

where $V_{new,k}$ tests only the capability newly added at M_k, and $K_{M,<k}=\prod_{j<k}K_{Mj}$ is the conjunction of all lower-M retention gates. Therefore a system may pass the M5-LONG lineage test and still fail M5 certification if any of M0-M4 is lost. Likewise, an MΩ candidate with a genuine Φ change cannot be certified if any M0-M5 retention test fails.

For MΩ specifically:

$$z_{M\Omega} = M_\Phi \cdot V_\Phi \cdot G_\Phi \cdot K_{M0:M5}$$

M_Φ establishes a genuine, in-scope, active and behaviorally relevant Φ0→Φ1 change; V_Φ establishes externally validated improvement; G_Φ guards migration, regression and resource confounds; $K_{M0:M5}$ enforces cumulative retention.

C. MΩ measurement note: an instrumented memory interface

Strong MΩ certification requires more than output improvement. The evaluator needs an instrumented interface I_Φ that snapshots D and Φ separately; records the Memory Manifest and Φ versions; verifies scope and activation; logs memory operations; supports matched controls/ablation and migration checks; and performs independent promotion/rollback. The tested agent, not the interface, must generate ΔΦ. A clean causal comparison holds D* fixed across A(D*,Φ0), A(D*,Φ1), and an ablation A(D*,Φ1−ΔΦ); hidden tasks, verifier state, thresholds, control assignment and promotion remain protected.

GUI-Grounded Cognitive Intelligence and GP Diagnostic Addendum (v7)

Status. This pre-ratification addendum operationally expands the Cognitive Intelligence measurement surface without changing the invariant meanings of C0–C6 or $C\Omega$. Computer-screen understanding becomes a first-class Cognitive Intelligence domain, while GUI Perception (GP0–GP5) remains a diagnostic perceptual subscale rather than a new top-level intelligence family or a replacement C ladder.

A. Method B: preserve C levels, add a GUI domain

The original C ladder continues to answer *how deep the cognition is*: reactive, contextual, compositional, strategic, expert, discovery, frontier-generalizing, and open-ended. GUI capability instead specifies *where that cognition is grounded*. We therefore define GUI-grounded witnesses

$$C_0^{\mathrm{GUI}} \to C_1^{\mathrm{GUI}} \to \cdots \to C_6^{\mathrm{GUI}} \to C_\Omega^{\mathrm{GUI}},$$

where C_k^{GUI} is a GUI/screen-domain witness of the existing semantic level C_k; it is not a new intelligence level.

The intended containment shorthand is

$$\boxed{\mathrm{GP} \subset C^{\mathrm{GUI}} \subset C, \qquad \mathrm{GP}_g \not\equiv C_g.}$$

The non-equivalence is essential. A model may accurately read and ground an unfamiliar screen yet fail strategic reasoning, or may reason strongly from an oracle screen representation while failing to perceive the native screenshot.

The common Cognitive domain panel is expanded to

$$\mathcal{D}_C = \{\text{general, science, mathematics, abstract, code, multimodal/spatial, GUI/screen, long-context}\}.$$

A score from one domain cannot silently substitute for another. Until ratification calibrates mandatory domain floors, systems lacking image/screen access report $C^{\mathrm{GUI}} = \mathrm{N/A}$ rather than zero; a broad multimodal-C claim must state whether GUI/screen is required.

B. GUI-grounded witnesses of the original C ladder

Witness	Original C construct	GUI-domain operational interpretation
C_0^{GUI}	Reactive	Read, identify, or localize one explicitly requested screen element with no hidden planning requirement.
C_1^{GUI}	Contextual	Infer the ordinary function of a visible interface element from screen context and a routine goal.
C_2^{GUI}	Compositional	Combine multiple GUI elements, states, spatial relations, panels, rows, or constraints to answer a structured question.
C_3^{GUI}	Strategic	Construct or select an interface-navigation/action strategy and transfer it to a held-out layout or unfamiliar application surface.
C_4^{GUI}	Expert	Reason robustly through a dense expert interface containing ambiguity, interacting constraints, and relevant distractors.
C_5^{GUI}	Discovery	Infer an unknown interface rule, workflow, transition model, or interaction mechanism and predict sealed GUI states.
C_6^{GUI}	Frontier-Generalizing	Integrate GUI understanding with several genuinely necessary domains, such as code, mathematics, science, documents, or spatial reasoning.
C_Ω^{GUI}	Open-Ended	Create a new interface representation, parser, abstraction, or cognitive method that causally expands the classes of GUI-grounded problems subsequently reachable.

C. GP0–GP5 is a diagnostic perceptual subscale

GP measures the fidelity and generality of the screen representation supplied to cognition. It must not be mapped one-to-one onto C levels.

GP	Capability	Operational criterion
GP0	Pixel/text/basic visual recognition	Recognizes visible text, simple icons, colors, and elementary screen features without claiming task understanding.
GP1	UI-element recognition	Distinguishes buttons, menus, fields, tabs, dialogs, lists, windows, and other interface objects.
GP2	Spatial grounding and hierarchy	Grounds a referred element to the correct region and represents containment, adjacency, ownership, labeling, and target relations.
GP3	Interface-state understanding	Identifies enabled/disabled, selected, focused, open/closed, loading, error, modal, and analogous states.
GP4	Dynamic GUI understanding	Tracks changes across screenshots/actions: what changed, whether the intended effect occurred, and what remains unresolved.
GP5	Novel-interface perceptual generalization	Grounds and interprets unfamiliar software/layouts without application-specific scripting or memorized element paths.

D. Failure decomposition and construct boundaries

A computer-use failure can arise from perception, reasoning, action execution, or delegation-process control. HIL should keep these causes legible rather than collapsing them into one score. A useful measurement decomposition is

$$\text{computer task outcome} \Leftarrow \text{GP} \ \wedge \ C^{\text{GUI}} \ \wedge \ \text{actuator/action fidelity} \ \wedge \ \text{DI process}.$$

Accordingly:

- **GP** measures screen perception/representation.
- C^{GUI} measures C-level reasoning grounded in a GUI.
- **Action fidelity** measures whether requested clicks, typing, scrolling, dragging, or other computer actions are executed correctly; it is not itself a C level.
- **DI** measures whether the complete GUI task can be delegated at task difficulty T, human cognitive intervention H, and reliability p.

Strong experiments therefore compare a native-screenshot arm with an oracle-screen-graph arm and, for end-to-end computer use, a native actuator with a perfect-actuator control. These interventions diagnose whether a failure is perceptual, cognitive/strategic, or motor/tool-execution limited. The same delivered-success event must not be counted twice as both C and DI evidence.

E. Reporting

A GUI-capable system should report at least

$$\boxed{C \text{ profile} \mid C^{\text{GUI}} \text{ domain level} \mid \text{GP level} \mid \text{GUI task } T/H/p.}$$

For example, a system may be globally $C4$ with C_3^{GUI}, GP5, and a computer-use delegation frontier of $T3/H1@p = 0.80$. These quantities answer different questions and should remain separately inspectable.